STAT 5200, Notes 2[1]

Minchul Shin[2]

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Readings. Read chapter 3 in W. The subject matter is asymptotic theory, and it is drawn from probability theory and statistical theory. The important topics are big oh and little oh notation, convergence in probability, convergence in distribution, Slutsky's theorem, the weak law of large numbers, the central limit theorem, the continuous mapping theorem, the Wald test, and the delta method.

Slutsky's theorem is misidentified in W. It is not W's Lemma 3.4, which is a version of continuous mapping theorem. One part of Slutsky's theorem is given by Lemma 3.7, which Wooldridge terms the asymptotic equivalence lemma.

Many of you will find this asymptotic theory to be heavy going, especially if it is relatively new to you. In this set of notes, I discuss the material, but do not dwell on it at great length. Some proofs are presented. I will be applying asymptotic theory throughout the course, and as we encounter it, I will explain how it is being used. So be patient, and don't get discouraged.

Big Oh and Little Oh. We discuss big oh and little oh notation for sequences. Our main interest is in use of the notation for sequences of random variables. However, to explain, we start by defining big oh and little oh notation for deterministic sequences. We begin by defining convergence and boundedness for a sequence of numbers.

Limit of a sequence of numbers. A sequence of numbers $\{a_n, n = 1, 2, ...\}$ converges to a limit a as $n \to \infty$ if, for any $\varepsilon > 0$, there exists an index value n_ε such that

$$|a_n - a| < \varepsilon \quad \text{for all} \quad n > n_\varepsilon$$

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[2]Email: mincshin@wharton.upenn.edu

We write

$$\lim_{n\to\infty} a_n = a.$$

Boundedness of a sequence of numbers. A sequence of numbers $\{a_n, n = 1, 2, ...\}$ is bounded if there exists a number b such that

$$|a_n| \leq b \quad \text{for } n = 1, 2, ...$$

Big oh for a sequence of numbers. A sequence of numbers $\{a_n, n = 1, 2, ...\}$ is $O(n^\lambda)$ if the sequence $n^{-\lambda} a_n$ is bounded. We say that the sequence is *at most of order* n^λ. If $\lambda = 0$, the sequence is bounded, and we write $a_n = O(1)$. We say that a_n is *big oh one.*

Little oh for a sequence of numbers. A sequence of numbers $\{a_n, n = 1, 2, ...,\}$ is $o(n^\lambda)$ if the sequence $n^{-\lambda} a_n$ tends to 0. If $\lambda = 0$, the sequence tends to 0, and we write $a_n = o(1)$. We say that a_n is *little oh one.*

If $a_n = o(1)$, it is also true that $a_n = O(1)$.

These definitions extend to sequences of vectors and matrices, if the results occur for each element of the vector or matrix.

Let's consider a few examples. (i) If $a_n = n(n+1)/2$, $a_n = O(n^2)$. We can also write $a_n = o(n^{2+\gamma})$ for all $\gamma > 0$. (ii) If $a_n = 1/n^{1/3}$, $a_n = O(n^{-1/3})$, and also $a_n = o(1)$. (iii) If $a_n = \log n^2$, $a_n = O(\log n)$, and also $a_n = o(n^\gamma)$ for all $\gamma > 0$.

Convergence in Probability, Boundedness in Probability. Let $\{x_n, n = 1, 2, ...,\}$ be a sequence of random variables. If

$$P\left(|x_n - a| > \varepsilon\right) \to 0$$

as $n \to \infty$ for any $\varepsilon > 0$, we say that the sequence *converges to a in probability*, and we write $x_n \to_p a$, or $\operatorname{plim} x_n = a$. We write $x_n = a + o_p(1)$, a plus little oh p one. If x is another random variable, $x_n \to_p x$ means the same as $x_n - x \to_p 0$.

The sequence $\{x_n, n = 1, 2, ...,\}$ is said to be *bounded in probability* if, for any $\varepsilon > 0$, there exists a constant $b_\varepsilon > 0$ and an integer n_ε such that

$$P\left(|x_n| \geq b_\varepsilon\right) < \varepsilon \quad \text{for } n \geq n_\varepsilon. \tag{1}$$

We write $x_n = O_p(1)$, *big oh p one.*

Lemma 3.1 in W states that $\{x_n, n = 1, 2, ...\}$ is bounded in probability if it converges in probability to a constant. That is, a $o_p(1)$ sequence is also $O_p(1)$.

Let's prove it. By the convergence in probability, it follows that, for any $\varepsilon > 0$, there is an integer n_ε such that

$$P(|x_n - a| > 1) < \varepsilon \quad \text{for } n \geq n_\varepsilon.$$

We can take b_ε in (1) to be $|a| + 1$, because, by the triangle inequality,

$$|x_n| = |x_n - a + a| \leq |x_n - a| + |a|,$$

so that

$$|x_n - a| \geq |x_n| - |a|, \text{ which implies } P(|x_n| > |a| + 1) = P(|x_n| - |a| > 1) \leq P(|x_n - a| > 1). \quad \square$$

The lemma also holds for sequences of vectors and matrices.

Now consider more generally the big oh and little oh notation for random variables.

Big oh and little oh for a sequence of random variables. The sequences $\{x_n, n = 1, 2, ...\}$ is $O_p(a_n)$ where $\{a_n, n = 1, 2, ...\}$ is a sequence of positive numbers, if $x_n/a_n = O_p(1)$. Similarly, the sequences $\{x_n, n = 1, 2, ...\}$ is $o_p(a_n)$ where $\{a_n, n = 1, 2, ...\}$ is a sequence of positive numbers, if $x_n/a_n = o_p(1)$. These are written as $x_n = O_p(a_n)$ and $o_p(a_n)$, respectively.

Common examples of these sequences are $O_p(n^{-1/2})$, $O_p(n^{-1})$, $o_p(n^{-1/2})$, and $o_p(n^{-1})$.

Often one has need to add or multiply big oh p one and little oh p one sequences. Some useful results are

1. $o_p(1) + o_p(1) = o_p(1)$
2. $O_p(1) + O_p(1) = O_p(1)$
3. $o_p(1) \cdot o_p(1) = o_p(1)$
4. $O_p(1) \cdot O_p(1) = O_p(1)$
5. $o_p(1) + O_p(1) = O_p(1)$
6. $o_p(1) \cdot O_p(1) = o_p(1)$

These definitions and properties can be applied to sequences of random vectors and random matrices, by establishing them element by element, as suggested above. For example, if $\{\mathbf{x}_n, n = 1, 2, ...\}$ is a sequence of $K \times 1$ random vectors and $\mathbf{a}$ is a constant $K \times 1$ vector, $\mathbf{x}_n \to_p \mathbf{a}$ if and only if the convergence occurs for each element. This is equivalent to

$$||\mathbf{x}_n - \mathbf{a}|| \to_p 0, \text{ where } ||\mathbf{c}|| = (\mathbf{c}'\mathbf{c})^{1/2} \text{ is the Euclidean length of the } K \times 1 \text{ vector } \mathbf{c}.$$

Also, if $\{\mathbf{X}_n, n = 1, 2, ...\}$ is a sequence of $K \times L$ random matrices and $\mathbf{A}$ is a constant $K \times L$ matrix, $\{\mathbf{X}_n \to_p \mathbf{A}\}$ if and only if the convergence occurs for each matrix element. This is equivalent to

$$||\mathbf{X}_n - \mathbf{A}|| \to_p 0, \text{ where } ||\mathbf{C}|| = (tr(\mathbf{C}'\mathbf{C}))^{1/2} \text{ is the Euclidean norm of the } K \times L \text{ matrix.}$$

Here $tr(\mathbf{B})$ denotes the trace (the sum of the diagonal elements) of the square matrix $\mathbf{B}$.

Lemma 3.3 in W gives an example of extension of the big oh and little oh results above to the vector and matrix case. It is a result we will need subsequently. I am modifying it here slightly.

Suppose $\{\mathbf{Z}_n, n = 1, 2, ...\}$ is a sequence of $J \times K$ random matrices for which $\mathbf{Z}_n = O_p(1)$, and $\mathbf{x}_n$ is a sequence of $K \times 1$ random vectors for which $\mathbf{x}_n = o_p(1)$. Then $\mathbf{Z}_n\mathbf{x}_n = O_p(1) \cdot o_p(1) = o_p(1)$.

The *continuous mapping theorem* for convergence in probability is Lemma 3.4 in W. It is widely applied. It states that the operation of limit in probability (convergence in probability) passes through continuous functions. (Recall that Lemma 3.4 is not Slutsky's theorem.)

Lemma 3.4. Suppose $\{\mathbf{x}_n, n = 1, 2, ...\}$ is a sequence of $K \times 1$ random vectors for which $\mathbf{x}_n \to_p \mathbf{c}$, a constant $K \times 1$ vector. Let $\mathbf{g}$ be a function from K-dimensional Euclidean space to J-dimensional Euclidean space that is continuous at $\mathbf{c}$. Then $\mathbf{g}(\mathbf{x}_n) \to_p \mathbf{g}(\mathbf{c})$.

Corollary 3.1 in W gives an important application of this lemma. Suppose $\{\mathbf{Z}_n, n = 1, 2, ...\}$ is a sequence of $K \times K$ random matrices for which $\mathbf{Z}_n \to_p \mathbf{A}$, where $\mathbf{A}$ is a $K \times K$ nonrandom invertible matrix. The result of the corollary is that

$$\mathbf{Z}_n^{-1} \to_p \mathbf{A}^{-1}.$$

However, the precise statement and the proof require some care, because some members of the sequence $\{\mathbf{Z}_n, n = 1, 2, ...\}$ may not be invertible. See the details in W. Note that the operation of matrix inversion is a continuous function on the space of invertible matrices.

Convergence in Distribution. The cumulative distribution function (cdf) of a (univariate) random variable X (I'll use capital letters to denote random variables and random vectors in this section) is the function F_X defined by

$$F_X(x) = P(X \le x), \quad -\infty < x < \infty.$$

The cdf F_X (i) is nondecreasing, (ii) satisfies

$$\lim_{x \to -\infty} F_X(x) = 0, \quad \lim_{x \to \infty} F_X(x) = 1,$$

and (iii) can be taken to be continuous from the right. Cdfs include three basic types, (i) those which are absolutely continuous (an absolutely continuous cdf is equal to the integral of its derivative – the derivative is the density), (ii) those which are pure jump functions (a pure jump function cdf has a probability mass function), and (iii) those which are singular (they have all their points of increase concentrated on a set of Lebesgue measure zero). Every cdf can be expressed as a combination of these three types,

$$F = c_1 F_1 + c_2 F_2 + c_3 F_3,$$

where $c_1 + c_2 + c_3 = 1$, F_1 is an absolutely continuous cdf, F_2 is a pure jump function cdf, and F_3 is a singular cdf.

If $\mathbf{X} = (X_1, ..., X_K)$ is a random vector in K dimensions, its cdf is defined by

$$F_X(\mathbf{x}) = P(X_1 \leq x_1, ..., X_K \leq x_K).$$

The properties given above for a univariate cdf can be extended to the multivariate case.

In this course, we will mostly, if not always, be concerned with a multivariate normal distribution. A multivariate normal distribution is absolutely continuous – it has a density. See Notes 1 for discussion of the bivariate normal distribution. The multivariate normal distribution is discussed generally later in these notes.

Now consider a sequence of (univariate) random variables, $\{X_n, n = 1, 2, ...\}$, and let F_n be the cdf of X_n. The sequence of random variables *converges in distribution* to the random variable X with cdf F if

$$F_n(x) \to F(x) \quad \text{as } n \to \infty$$

at all points x where F is continuous. This convergence is denoted by

$$X_n \to_d X.$$

Now consider the multivariate case. Let $\{\mathbf{X}_n, n = 1, 2, ...\}$ be a sequence of K-dimensional random vectors. The sequence converges in distribution to the K-dimensional random vector $\mathbf{X}$ if and only if

$$\mathbf{c}'\mathbf{X}_n \to_d \mathbf{c}'\mathbf{X}$$

for any constant $K \times 1$ vector $\mathbf{c}$. This convergence is denoted by

$$\mathbf{X}_n \to_d \mathbf{X}.$$

The use of linear combinations $\mathbf{c}'\mathbf{X}_n$ of the components of $\mathbf{X}_n$ is known as the Cramér-Wold device. It allows one to reduce the problem of convergence in distribution in the multivariate case to convergence in distribution in the univariate case. Underlying this result is the continuity theorem for characteristic functions (which I may discuss briefly in class). Note that, in contrast to convergence in probability, convergence in distribution for sequences of random vectors and random matrices does not follow from convergence in distribution element by element. All possible linear combinations of the elements must converge in distribution.

If the limit $\mathbf{X}$ has the multivariate normal distribution with $\mathbf{m}$ and covariance matrix $\mathbf{V}$, say that $\mathbf{X}_n$ has a limiting normal distribution with mean $\mathbf{m}$ and covariance matrix $\mathbf{V}$. We write

$$\mathbf{X}_n \rightarrow_d N(\mathbf{m}, \mathbf{V}), \quad \text{or} \quad \mathbf{X}_n \sim_a N(\mathbf{m}, \mathbf{V}),$$

where for the latter notation we say that $\mathbf{X}_n$ is asymptotically normal with mean $\mathbf{m}$ and covariance matrix $\mathbf{V}$.

There are some connections between big oh p, convergence in probability, and convergence in distribution.

- We have already noted Wooldridge's Lemma 3.1, which states that if $\mathbf{X}_n \rightarrow_p \mathbf{a}$, a constant vector, then $\mathbf{X}_n = O_p(1)$.
- Lemma 3.5 in W states that if $\mathbf{X}_n \rightarrow_d \mathbf{X}$, then $\mathbf{X}_n = O_p(1)$. We will have occasion to apply this result.
- Convergence in probability $\mathbf{X}_n \rightarrow_p \mathbf{X}$ implies convergence in distribution $\mathbf{X}_n \rightarrow_d \mathbf{X}$. A proof for the scalar case is in the Appendix to these notes. The converse of this does not hold unless the convergence in distribution is to a constant (when the convergence is to a degenerate random variable). Note that the converse doesn't make much sense, because convergence in probability requires knowledge of the joint distribution of $\mathbf{X}_n$ and $\mathbf{X}$, whereas convergence in distribution requires only knowledge of the marginal distributions of $\mathbf{X}_n$ and $\mathbf{X}$.

The continuous mapping theorem for convergence in distribution is Lemma 3.6 in W. If $\{\mathbf{X}_n, n = 1, 2, ...\}$ is a sequence of $K \times 1$ random vectors for which $\mathbf{X}_n \rightarrow_d \mathbf{X}$, and $\mathbf{g}$ is a function from K-dimensional Euclidean space to J-dimensional Euclidean space that is continuous, then

$$\mathbf{g}(\mathbf{X}_n) \rightarrow_d \mathbf{g}(\mathbf{X}).$$

In other words, convergence in distribution passes through continuous functions.

Let's consider some examples of use of the continuous mapping theorem which we will encounter. Before doing so, let's review details and notation for normal and chi-square distributions. In Notes 1 there was some discussion of bivariate and multivariate normal distributions.

The normal distribution. If X is normally distributed with mean μ and variance σ^2, we write that X is $N(\mu, \sigma^2)$. The density function (the distribution is absolutely continuous) is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

For constants a and b, the distribution of $aX + b$ is also normal, specifically, it is

$$N(a\mu + b, a^2\sigma^2).$$

The distribution of

$$Z = \frac{X - \mu}{\sigma} \tag{2}$$

is therefore $N(0,1)$, termed *standard normal.*

In the multivariate case, the $K \times 1$ vector $\mathbf{X}$ is normally distributed with mean vector μ and (assume positive definite) covariance matrix Σ, written $N(\mu, \Sigma)$, if it has density function

$$f_X(\mathbf{x}) = \frac{1}{(2\pi)^{K/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)\right), \quad \mathbf{x} \in I\!R^K,$$

where $|\Sigma|$ is the determinant of Σ, and $I\!R^K$ denotes K-dimensional Euclidean space. If $\mathbf{A}$ is a $Q \times K (Q \le K)$ matrix of constants and $\mathbf{b}$ is a $Q \times 1$ vector of constants, the distribution of $\mathbf{AX}+\mathbf{b}$ is Q-dimensional normal, in particular,

$$N(\mathbf{A}\mu + \mathbf{b},\ \mathbf{A}\Sigma\mathbf{A}').$$

That is, linear transformations of normal vectors are also normal. It follows that the distribution of

$$\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \mu) \tag{3}$$

is $N(\mathbf{0}, \mathbf{I})$. Compare (2). I'll discuss $\Sigma^{1/2}$ in class.

The chi-square distribution. If $X_1, ..., X_n$ are i.i.d., $N(0,1)$, then

$$Y = X_1^2 + \cdots + X_n^2$$

has a chi-square distribution with n degrees of freedom (df), denoted χ_n^2. The density is

$$f_Y(y) = \frac{1}{2^{\frac{1}{2}n}\Gamma(\frac{1}{2}n)} y^{\frac{1}{2}n-1} \exp\left(-\frac{1}{2}y\right), \quad 0 < y < \infty.$$

The function

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}\exp(-x)dx$$

is defined for $\alpha > 0$. It is called the gamma function. Using integration by parts, one can show that

$$\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1), \quad \alpha > 1.$$

Moreover, with a change of variables and use of the normal density function, one can show that $\Gamma(1/2) = \sqrt{\pi}$. The mean and variance of the chi-square distribution with n df are n and $2n$, respectively.

By the way, if $\mathbf{Z}$ is $N(\mathbf{0}, \mathbf{I})$, a necessary and sufficient condition for the quadratic form

$$\mathbf{Z}'\mathbf{A}\mathbf{Z}$$

to have a chi-square distribution is that $\mathbf{A}$ be idempotent (without loss of generality $\mathbf{A}$ is symmetric). ($\mathbf{A}$ is idempotent if $\mathbf{A}^2 = \mathbf{A}$.) When this is the case, the degrees of freedom of the chi-square distribution are equal to the rank of $\mathbf{A}$. In this case (symmetric idempotent $\mathbf{A}$), the rank of $\mathbf{A}$ is also equal to the trace of $\mathbf{A}$. The eigenvalues of a symmetric idempotent matrix are all 0 or 1.

Examples (use of the continuous mapping theorem).

(1) If $\{x_n, n = 1, 2, ...\}$ is a sequence of random variables for which $x_n \to_d N(0, v)$, then $x_n^2/v \to_d \chi_1^2$.

(2) Let $\{\mathbf{X}_n, n = 1, 2, ...\}$ be a sequence of $K \times 1$ random vectors for which $\mathbf{X}_n \to_d N(\mathbf{0}, \mathbf{V})$, where $\mathbf{V}$ is a full-rank matrix, and assume $\mathbf{A}$ is a $K \times Q$ nonrandom matrix. Then

   (i) $\mathbf{A}'\mathbf{X}_n \to_d N(\mathbf{0}, \mathbf{A}'\mathbf{V}\mathbf{A})$, a Q-dimensional normal distribution, and

   (ii) $\mathbf{X}_n'\mathbf{V}^{-1}\mathbf{X}_n \to_d \chi_K^2$.

Slutsky's Theorem. Slutsky's results were published in 1925 in the journal *Metron* (in German). I'll begin by stating the theorem as it is often quoted. Then I'll give a more compact form of the statement.

Let $\{\mathbf{X}_n, n = 1, 2, ...\}$ and $\{\mathbf{Y}_n, n = 1, 2, ...\}$ be sequences of $K \times 1$ random vectors for which $\mathbf{X}_n \to_d \mathbf{X}$ and $\mathbf{Y}_n \to_p \mathbf{a}$, where $\mathbf{a}$ is a constant vector. Then the following holds:

(i) $\mathbf{X}_n + \mathbf{Y}_n \to_d \mathbf{X} + \mathbf{a}$

Now let $\{\mathbf{Y}_n, n = 1, 2, ...,\}$ be a sequence of $Q \times K$ random matrices such that $\mathbf{Y}_n \to_p \mathbf{A}$, where $\mathbf{A}$ is a constant $Q \times K$ matrix, and as before, assume $\mathbf{X}_n \to_d \mathbf{X}$. Then

(ii) $\mathbf{Y}_n\mathbf{X}_n \to_d \mathbf{AX}$.

Assume also that $Q = K$ and the matrices $\mathbf{Y}_n$ and $\mathbf{A}$ are invertible. Then,

(iii) $\mathbf{Y}_n^{-1}\mathbf{X}_n \to_d \mathbf{A}^{-1}\mathbf{X}$.

We will apply this theorem extensively. Let's give a very brief summary of the proof of (i). From the given assumptions, the vector $(\mathbf{X}_n, \mathbf{Y}_n)$ converges jointly in distribution to $(\mathbf{X}, \mathbf{a})$ (the proof of this statement does require some care). Then the result follows from the continuous mapping theorem. The proofs of (ii) and (iii) follow similar steps.

With the above assumptions, Slutsky's theorem may be stated as follows. Let $\mathbf{g}$ be a continuous function. Then,

$$\mathbf{g}(\mathbf{X}_n, \mathbf{Y}_n) \to_d \mathbf{g}(\mathbf{X}, \mathbf{A}).$$

The Weak Law of Large Numbers. The weak law of large numbers states that, loosely speaking, averages converge in probability to their expected values. The precise statement is the following.

Suppose $\{\mathbf{X}_i, i = 1, 2, ...\}$ is a sequence of independent $K \times 1$ random vectors, all with the same distribution, that is, independent and identically distributed (i.i.d.). Moreover, assume the mean of the random vectors is defined,

$$\mu = E[\mathbf{X}_i].$$

Then

$$\frac{1}{N}\sum_{i=1}^{N} \mathbf{X}_i \to_p \mu.$$

The Central Limit Theorem. This is the Lindeberg–Lévy version of the central limit theorem. Suppose $\{\mathbf{X}_i, i = 1, 2, ...\}$ is a sequence of iid $K \times 1$ random vectors. Moreover, assume

$$E[\mathbf{X}_i] = \mathbf{0} \quad \text{and} \quad E[\mathbf{X}_i\mathbf{X}_i'] = \mathbf{B}.$$

That is, the vectors have mean vector zero and covariance matrix $\mathbf{B}$. Then

$$\frac{1}{\sqrt{N}}\sum_{i=1}^{N} \mathbf{X}_i \to_d N(\mathbf{0}, \mathbf{B}).$$

Note the division of the sum is by $\sqrt{N}$, but for the law of large numbers, division of the sum is by N.

Behavior of Estimators. Suppose $\{\hat{\theta}_N, N = 1, 2, ...\}$ is a sequence of estimators of the $P \times 1$ vector parameter $\theta \in \Theta$, arising from a sample of size N (N observations). If

$$\hat{\theta}_N \to_p \theta,$$

$\hat{\theta}_N$ is said to be a *consistent estimator* of θ. Furthermore, if

$$\sqrt{N}\left(\hat{\theta}_N - \theta\right) \to_d N(\mathbf{0}, \mathbf{V}), \tag{4}$$

where $\mathbf{V}$ is $P \times P$ and positive semidefinite, we say that $\hat{\theta}_N$ is *asymptotically normally distributed* and that $\sqrt{N}(\hat{\theta}_N - \theta)$ has *asymptotic covariance matrix* $\mathbf{V}$. In this case, it is common to treat $\hat{\theta}_N$ as having a normal distribution with mean θ and covariance matrix $\mathbf{V}/N$, when N is large. One writes

$$\hat{\theta}_N \sim N(\theta, \mathbf{V}/N)$$

and refers to $\mathbf{V}/N$ as the asymptotic variance (asymptotic covariance matrix) of $\widehat{\theta}_N$,

$$Avar(\widehat{\theta}_N) = \mathbf{V}/N.$$

However, in practice we do not know $\mathbf{V}$, and we estimate it from the data. Let $\widehat{\mathbf{V}}_N$ be a consistent estimator of $\mathbf{V}$, $\widehat{\mathbf{V}}_N \to_p \mathbf{V}$. Then, the estimator of $Avar(\widehat{\theta}_N) = \mathbf{V}/N$ is

$$\widehat{Avar}(\widehat{\theta}_N) = \widehat{\mathbf{V}}_N/N.$$

The *asymptotic standard errors* of the elements of $\widehat{\theta}_N$ are the square roots of the diagonal entries of the matrix $\widehat{\mathbf{V}}_N/N$. That is, if $\widehat{\theta}_{jN}$ is the j-th element of $\widehat{\theta}_N$ and $\widehat{v}_{jjN}$ is the j-th diagonal element of $\widehat{\mathbf{V}}_N$, the standard error of $\widehat{\theta}_{jN}$ is

$$se(\widehat{\theta}_{jN}) = \left(\widehat{v}_{jjN}/N\right)^{1/2}.$$

If (4) holds, it follows by Lemma 3.5 in W that

$$\sqrt{N}(\widehat{\theta}_N - \theta) = O_p(1), \quad \text{or} \quad \widehat{\theta}_N - \theta = O_p(N^{-1/2}). \tag{5}$$

When (5) holds, we say that $\widehat{\theta}_N$ is a $\sqrt{N}$-*consistent estimator* of θ. This statement tells us more than that $\widehat{\theta}_N$ is consistent for θ – it specifies that the rate of convergence is given by the square root of the sample size N.

The Wald Test Statistic. The Wald test is described in Lemma 3.8 in W. Suppose $\widehat{\theta}_N$ is an estimator of the $P \times 1$ vector parameter θ, arising from a sample of size N, and assume that

$$\sqrt{N}(\widehat{\theta}_N - \theta) \text{ is asymptotically } N(\mathbf{0}, \mathbf{V}).$$

Let $\mathbf{R}$ be a $Q \times P$ matrix of constants, $Q \leq P$, with row rank Q. Then,

$$\sqrt{N}\mathbf{R}(\widehat{\theta}_N - \theta) \text{ is asymptotically } N(\mathbf{0}, \mathbf{RVR}').$$

It follows by the continuous mapping theorem that

$$\left[\sqrt{N}\mathbf{R}(\widehat{\theta}_N - \theta)\right]' (\mathbf{RVR}')^{-1} \left[\sqrt{N}\mathbf{R}(\widehat{\theta}_N - \theta)\right] \text{ is asymptotically chi-square with } Q \text{ df.}$$

Furthermore, if $\widehat{\mathbf{V}}_N$ is a consistent estimator of $\mathbf{V}$, by Slutsky's theorem,

$$\left[\sqrt{N}\mathbf{R}(\widehat{\theta}_N - \theta)\right]' (\mathbf{R}\widehat{\mathbf{V}}_N\mathbf{R}')^{-1} \left[\sqrt{N}\mathbf{R}(\widehat{\theta}_N - \theta)\right] = \left[\mathbf{R}(\widehat{\theta}_N - \theta)\right]' (\mathbf{R}(\widehat{\mathbf{V}}_N/N)\mathbf{R}')^{-1} \left[\mathbf{R}(\widehat{\theta}_N - \theta)\right] \tag{6}$$

is asymptotically chi-square with Q df. Expression (6) is a Wald statistic. It is common to divide (6) by Q and treat it as an F-statistic. I'll discuss this in class and in later notes.

The Delta Method. Suppose $\widehat{\theta}_N$ is an estimator of a $P \times 1$ vector parameter $\theta = (\theta_1, ..., \theta_P)'$, and assume that $\widehat{\theta}_N$ is $\sqrt{N}$-asymptotically normally distributed. That is, $\sqrt{N}(\widehat{\theta}_N - \theta)$ converges in distribution, as N tends to infinity, to $N(\mathbf{0}, \mathbf{W})$, where $\mathbf{W}$ is a $P \times P$ covariance matrix. In other words, for large N we can treat $\widehat{\theta}_N$ as $N(\theta, \mathbf{W}/N)$. We denote the estimator of $\mathbf{W}/N$ by $\widehat{\mathbf{W}}/N$.

Next, suppose that $\mathbf{c}(\theta) = (c_1(\theta), ..., c_Q(\theta))'$ is a $Q \times 1$ vector defining a transformation of the parameter θ. Thus, there are Q scalar transformations of the vector parameter θ. We assume that $Q \leq P$, and that the transformation has continuous first-order partial derivatives. Form the $Q \times P$ matrix of partial derivatives

$$\mathbf{C}(\theta) = \left(\partial c_i(\theta)/\partial\theta_j, \quad i = 1, ..., Q, j = 1, ..., P\right).$$

How do we estimate $\mathbf{c}(\theta)$, and what is the asymptotic distribution of the estimator? To estimate $\mathbf{c}(\theta)$ we use $\mathbf{c}(\widehat{\theta}_N)$, a plug-in estimator. As N tends to infinity, one can prove that

$$\sqrt{N}\left(\mathbf{c}(\widehat{\theta}_N) - \mathbf{c}(\theta)\right)$$

converges in distribution to

$$N(\mathbf{0},\ \mathbf{C}(\theta)\mathbf{W}\mathbf{C}(\theta)').$$

In other words, for large N we can treat $\mathbf{c}(\widehat{\theta}_N)$ as

$$N(\mathbf{c}(\theta),\ \mathbf{C}(\theta)\mathbf{W}\mathbf{C}(\theta)'/N).$$

We estimate $\mathbf{C}(\theta)\mathbf{W}\mathbf{C}(\theta)'/N$ by $\mathbf{C}(\widehat{\theta}_N)\widehat{\mathbf{W}}\mathbf{C}(\widehat{\theta_N})'/N$.

The proof of this result uses the mean value theorem from calculus and Slutsky's theorem. See pages 46–47 in W.

To find asymptotic standard errors and asymptotic distributions, the delta method is not needed in our present context of estimation if the rows of $\mathbf{c}(\theta)$ define *linear* functions of the parameter vector θ. (The delta method will still give correct results for linear functions.) Let's briefly look at the case where $\mathbf{c}(\theta)$ is defined by linear functions of the parameter vector. We suppose $\mathbf{R}$ is a $Q \times P$ matrix of constants and write $\mathbf{r}(\theta) = \mathbf{R}\theta$. We estimate $\mathbf{r}(\theta)$ by $\mathbf{r}(\widehat{\theta}_N) = \mathbf{R}\widehat{\theta}_N$. Then for large N we treat the distribution of $\mathbf{r}(\widehat{\theta}_N)$ as $N(\mathbf{r}(\theta, \mathbf{RWR}'/N)$, and we estimate $\mathbf{RWR}'/N$ by $\mathbf{R}\widehat{\mathbf{W}}\mathbf{R}'/N$. See Lemmas 3.6 and 3.8 in W.

For nonlinear functions of the parameter vector, though, the delta method can be employed to obtain asymptotic standard errors and asymptotic distributions. Alternatives to the delta method for standard error calculation are the bootstrap and the jackknife, which we will consider later.

Appendix. Recall one of the claims in the notes regarding the relationship between "converges in probability" and "converges in distribution",

THEOREM. If $\mathbf{X}_n \to_p \mathbf{X}$, then $\mathbf{X}_n \to_d \mathbf{X}$.

We prove this for the scalar case. To begin, we prove a lemma.

LEMMA. Let X and Y be random variables, a a constant, and $\varepsilon > 0$. Then,

$$P(Y \leq a) \leq P(X \leq a + \varepsilon) + P(|Y - X| > \varepsilon).$$

Proof of Lemma.

$$\begin{aligned}
P(Y \leq a) &= P(Y \leq a, X \leq a + \varepsilon) + P(Y \leq a, X > a + \varepsilon) \\
&\leq P(X \leq a + \varepsilon) + P(Y - X \leq a - X, a - X < -\varepsilon) \\
&\leq P(X \leq a + \varepsilon) + P(Y - X < -\varepsilon) \\
&\leq P(X \leq a + \varepsilon) + P(Y - X < -\varepsilon) + P(Y - X > \varepsilon) \\
&= P(X \leq a + \varepsilon) + P(|Y - X| > \varepsilon).
\end{aligned}$$

□

Proof of Theorem.

Let F_n denote the cdf of X_n and F the cdf of X, let a be a continuity point of F, and $\varepsilon > 0$. By the lemma,

$$P(X_n \leq a) \leq P(X \leq a + \varepsilon) + P(|X_n - X| > \varepsilon) \tag{A.1}$$

and

$$P(X \leq a - \varepsilon) \leq P(X_n \leq a) + P(|X_n - X| > \varepsilon). \tag{A.2}$$

Then, by (A.1) and (A.2),

$$P(X \leq a - \varepsilon) - P(|X_n - X| > \varepsilon) \leq P(X_n \leq a) \leq P(X \leq a + \varepsilon) + P(|X_n - X| > \varepsilon). \tag{A.3}$$

Take the limit as $n \to \infty$ in (A.3). The result is

$$F(a - \varepsilon) - 0 \leq \lim_{n \to \infty} F_n(a) \leq F(a + \varepsilon) + 0. \tag{A.4}$$

Finally, take the limit as $\varepsilon \to 0$ in (A.4). $\square$

Positive semidefinite and positive definite matrices. Let $\mathbf{A}$ be a $K \times K$ matrix with real-valued entries and suppose that $\mathbf{c}$ is a $K \times 1$ vector. $\mathbf{A}$ is defined to be positive semidefinite if $\mathbf{c}'\mathbf{A}\mathbf{c} \geq 0$ for any choice of $\mathbf{c}$. Further, $\mathbf{A}$ is positive definite if $\mathbf{c} = 0$ is the only choice for which $\mathbf{c}'\mathbf{A}\mathbf{c} = 0$.

Also note that without loss of generality we can assume that $\mathbf{A}$ is a symmetric matrix.

A positive definite matrix has full rank. The eigenvalues of a positive definite matrix are all positive.

Covariance matrices are positive semidefinite

Suppose $\mathbf{X}$ is a $K \times 1$ random vector with covariance matrix Σ, defined by

$$\Sigma = cov(\mathbf{X}) = E\left[(\mathbf{X} - E[\mathbf{X}])(\mathbf{X} - E[\mathbf{X}])'\right].$$

Without loss of generality we can assume that the mean vector $E[\mathbf{X}] = \mathbf{0}$ and write

$$\Sigma = cov(\mathbf{X}) = E[\mathbf{X}\mathbf{X}'].$$

Now let $\mathbf{c}$ be a $K \times 1$ vector of constants, and consider the variance of $\mathbf{c}'\mathbf{X}$. We have, for any choice of $\mathbf{c}$,

$$0 \leq var(\mathbf{c}'\mathbf{X}) = E[\mathbf{c}'\mathbf{X}\mathbf{X}'\mathbf{c}] = \mathbf{c}'E[\mathbf{X}\mathbf{X}']\mathbf{c} = \mathbf{c}'\Sigma\mathbf{c},$$

where, without loss of generality, we have assumed $E[\mathbf{X}] = \mathbf{0}$.

Square root of a positive definite matrix. Let $\mathbf{A}$ be a positive definite matrix. Perform an eigenvector, eigenvalue decomposition, writing

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1},$$

where the columns of $\mathbf{Q}$ are eigenvectors of $\mathbf{A}$ and $\mathbf{\Lambda}$ is a diagonal matrix with entries equal to the eigenvalues of $\mathbf{A}$. The eigenvalues are all real numbers. Positive definiteness implies that the diagonal elements of $\mathbf{\Lambda}$ are all positive. Next we can write

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^{-1}\mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^{-1}.$$

Thus, the square root of $\mathbf{A}$ is $\mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^{-1}$. This proof is also valid if $\mathbf{A}$ is positive semidefinite.

Eigenvalues of an idempotent matrix. Suppose $\mathbf{A}$ is idempotent, that is, $\mathbf{A}^2 = \mathbf{A}$. Write

$$\mathbf{\Lambda} = \mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \mathbf{Q}^{-1}\mathbf{A}^2\mathbf{Q} = \mathbf{Q}^{-1}\mathbf{A}\mathbf{Q}\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \mathbf{\Lambda}^2.$$

The only possible eigenvalue solutions are 0 and 1.